

- 4** Redwood trees are the tallest living organisms on Earth. Some of the older trees are more than 2000 years old.

Dendrochronology can be used to work out how old a tree is and how much it has grown each year.

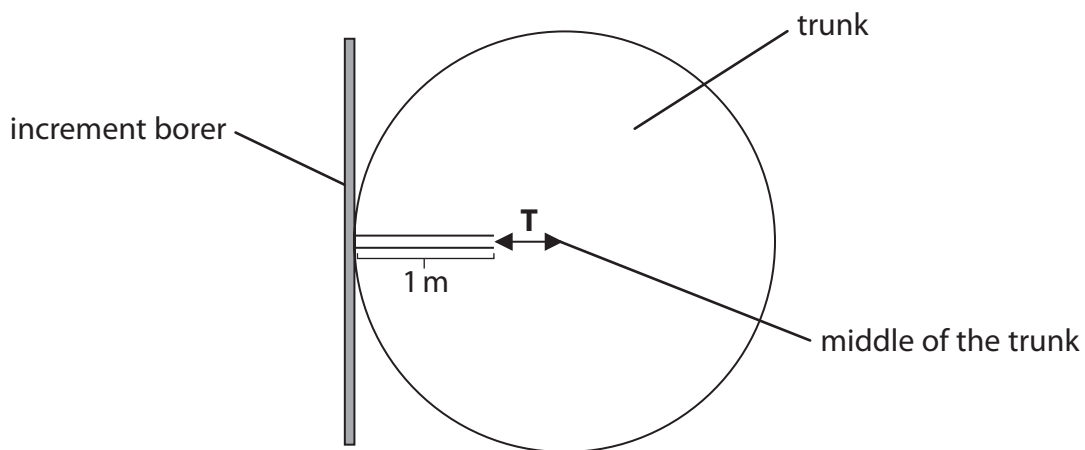
A sample of the tree trunk core can be taken using an increment borer.

The photograph shows an increment borer being used to take a sample of a tree trunk core.



(Source: © Custom Life Science Images / Alamy Stock Photo)

- (a) The diagram shows an increment borer pushed into the trunk of a large Redwood tree.



An increment borer 1 m long was inserted into a Redwood tree. The borer did not reach the middle of the trunk. The distance from the end of the borer to the middle of the trunk is **T** cm.

- (i) Calculate the radius (r) of a tree with a circumference (C) of 8 m.

Use the formula: $r = \frac{C}{2\pi}$

Give your answer to **two** decimal places.

(1)

Answer m

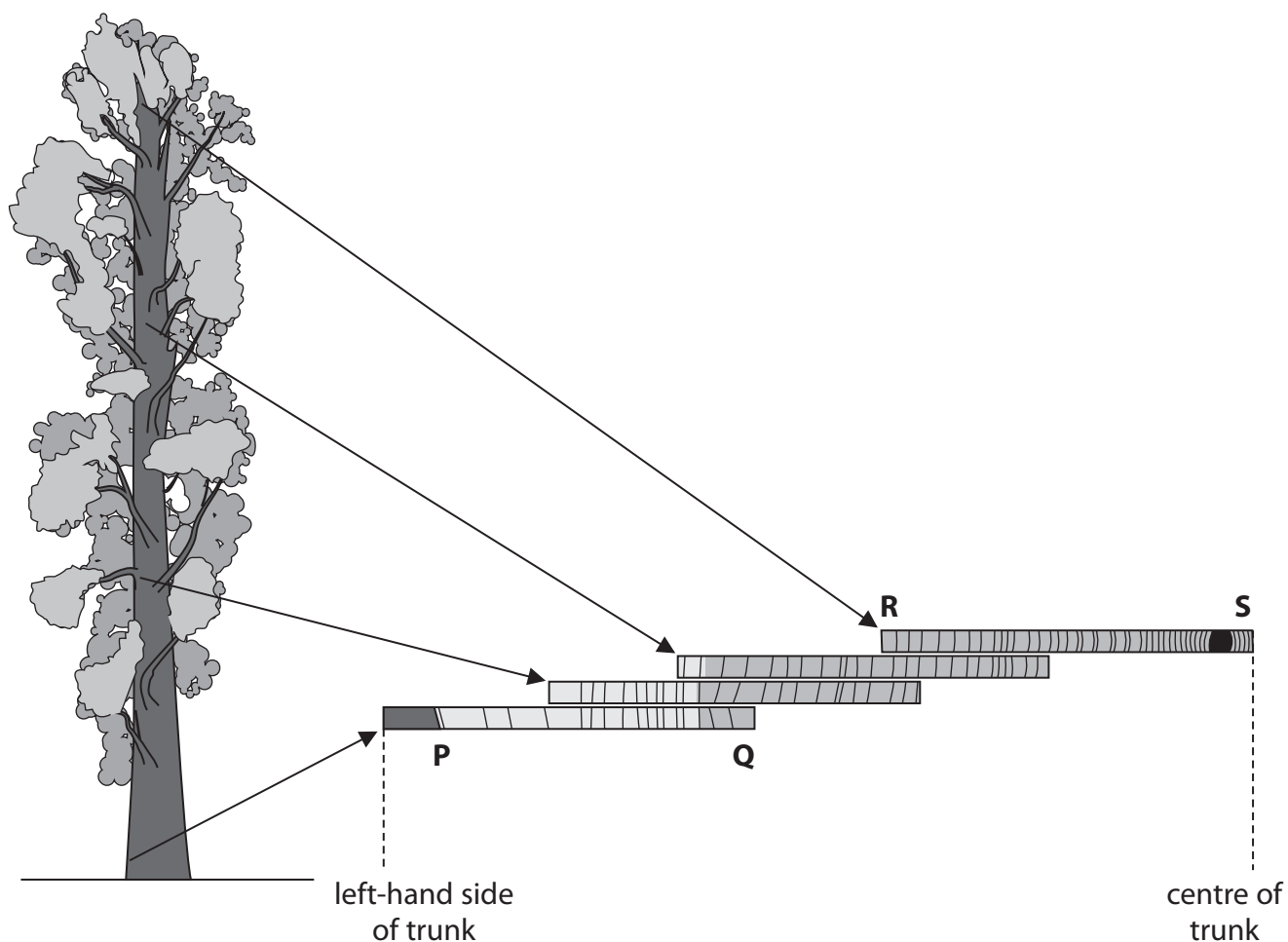
- (ii) Calculate the distance **T** as a percentage of this radius.

(1)

Answer %

- (b) To calculate how old a tree is and how much it has grown, core samples are taken at different heights up the tree.

These core samples are then aligned by matching the rings, as shown in the diagram.



- (i) Which row of the table shows the newest and oldest rings in this tree?

(1)

		newest ring	oldest ring
☐	A	P	Q
☐	B	P	S
☐	C	S	R
☐	D	S	P

(ii) Explain how these core samples can be used to calculate the age of this tree.

Use the information in the question and the diagram to support your answer.

(3)

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(iii) Describe how the growth rate of this tree can be calculated.

(2)

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(Total for Question 4 = 8 marks)